

**TUGAS PRAKTIK KOMPREHENSIF**  
**PRAKTIKUM 1 – SISTEM OPERASI**



Dosen pengampu  
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Disusun Oleh :  
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## Tugas 1: Process Manager

```
yoga2410651057@MSI: ~  
divoabdil@MSI:~$ sudo usermod -aG sudo yoga2410651057  
divoabdil@MSI:~$ logout  
yoga2410651057@MSI:~$ nano process_manager.c  
yoga2410651057@MSI:~$ gcc -o process_manager process_manager.c  
yoga2410651057@MSI:~$ ./process_manager  
Child 1: Masukkan angka untuk faktorial: Child 2: Masukkan batas bilangan prima: Child 3: Masukkan string untuk dibalik: 5  
Child 1 PID: 2996, Exit status: 0, Waktu: 9s -755321us, Faktorial: 120  
^C  
yoga2410651057@MSI:~$ /tmp/monitor_dir  
-bash: /tmp/monitor_dir: No such file or directory  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
yoga2410651057@MSI:~$ ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:42 /tmp/monitor_dir  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:42 /tmp/monitor_dir  
yoga2410651057@MSI:~$ nano monitor.c  
yoga2410651057@MSI:~$ gcc monitor.c -o monitor  
yoga2410651057@MSI:~$ ./monitor  
/tmp/monitor_dir  
touch /tmp/monitor_dir/test1.txt  
^C  
yoga2410651057@MSI:~$ touch /tmp/monitor_dir/test1.txt  
yoga2410651057@MSI:~$ echo "Hello" > /tmp/monitor_dir/test1.txt  
yoga2410651057@MSI:~$ tmp/monitor_dir  
-bash: tmp/monitor_dir: No such file or directory  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
yoga2410651057@MSI:~$ ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:45 /tmp/monitor_dir  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:45 /tmp/monitor_dir  
yoga2410651057@MSI:~$ nano monitor.c  
yoga2410651057@MSI:~$ gcc monitor.c -o monitor  
yoga2410651057@MSI:~$ ./monitor  
^[[T  
kill -SIGTERM <PID_monitor>  
^C  
yoga2410651057@MSI:~$ cat /tmp/monitor_log.txt  
yoga2410651057@MSI:~$ ^C  
yoga2410651057@MSI:~$ ^C  
yoga2410651057@MSI:~$ while(1) {  
    printf("myshell> ");
```

## Tugas 2: File Monitor

```
yoga2410651057@MSI: ~  
divoabdil@MSI:~$ sudo usermod -aG sudo yoga2410651057  
divoabdil@MSI:~$ logout  
yoga2410651057@MSI:~$ nano process_manager.c  
yoga2410651057@MSI:~$ gcc -o process_manager process_manager.c  
yoga2410651057@MSI:~$ ./process_manager  
Child 1: Masukkan angka untuk faktorial: Child 2: Masukkan batas bilangan prima: Child 3: Masukkan string untuk dibalik: 5  
Child 1 PID: 2996, Exit status: 0, Waktu: 9s -755321us, Faktorial: 120  
^C  
yoga2410651057@MSI:~$ /tmp/monitor_dir  
-bash: /tmp/monitor_dir: No such file or directory  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
yoga2410651057@MSI:~$ ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:42 /tmp/monitor_dir  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:42 /tmp/monitor_dir  
yoga2410651057@MSI:~$ nano monitor.c  
yoga2410651057@MSI:~$ gcc monitor.c -o monitor  
yoga2410651057@MSI:~$ ./monitor  
/tmp/monitor_dir  
touch /tmp/monitor_dir/test1.txt  
^C  
yoga2410651057@MSI:~$ touch /tmp/monitor_dir/test1.txt  
yoga2410651057@MSI:~$ echo "Hello" > /tmp/monitor_dir/test1.txt  
yoga2410651057@MSI:~$ tmp/monitor_dir  
-bash: tmp/monitor_dir: No such file or directory  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
yoga2410651057@MSI:~$ ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:45 /tmp/monitor_dir  
yoga2410651057@MSI:~$ mkdir -p /tmp/monitor_dir  
ls -ld /tmp/monitor_dir  
drwxrwxr-x 2 yoga2410651057 yoga2410651057 4096 Oct 17 00:45 /tmp/monitor_dir  
yoga2410651057@MSI:~$ nano monitor.c  
yoga2410651057@MSI:~$ gcc monitor.c -o monitor  
yoga2410651057@MSI:~$ ./monitor  
^[[T  
kill -SIGTERM <PID_monitor>  
^C  
yoga2410651057@MSI:~$ cat /tmp/monitor_log.txt  
yoga2410651057@MSI:~$ ^C  
yoga2410651057@MSI:~$ ^C  
yoga2410651057@MSI:~$ while(1) {  
    printf("myshell> ");
```

## Tugas 3: Custom Shell

```
yoga2410651057@MSI:~$ nano myshell.c
yoga2410651057@MSI:~$ gcc myshell.c -o myshell
myshell.c: In function 'main':
myshell.c:12:13: warning: implicit declaration of function 'strncmp' [-Wimplicit-function-declaration]
   12 |     if (strncmp(input, "exit", 4) == 0)
       |         ^
myshell.c:3:1: note: include '<string.h>' or provide a declaration of 'strncmp'
   2 | #include <stdlib.h>
+++ |+#include <string.h>
   3 |
myshell.c:12:36: warning: 'strncmp' argument 3 type is 'int' where 'long unsigned int' is expected in a call to built-in function declared without prototype [-Wbuiltin-declaration-mismatch]
   12 |     if (strncmp(input, "exit", 4) == 0)
       |                            ^
<built-in>: note: built-in 'strncmp' declared here
yoga2410651057@MSI:~$ ./myshell
myshell> mau
Kamu mengetik: mau
myshell> ^C
yoga2410651057@MSI:~$ nano myshell.c
yoga2410651057@MSI:~$ gcc myshell.c -o myshell
yoga2410651057@MSI:~$ ./myshell
myshell> pwd
/home/yoga2410651057
myshell> echo Hello World
Hello World
myshell> ls | grep .c
monitor.c
myshell.c
process_manager
process_manager.c
myshell> ls > output.txt
ls: cannot access '>': No such file or directory
ls: cannot access 'output.txt': No such file or directory
myshell> ls
monitor monitor.c myshell myshell.c process_manager process_manager.c
myshell> cat
cat output.txt
cat output.txt
sleep 10 &
sleep 10 &
```

## Tugas 4: System Information Tool

```
divo@MSI:~$ cat /proc/meminfo
Mem: total: 32768 kB, used: 10240 kB, free: 22528 kB, shared: 0 kB, buff/cache: 6024 kB, available: 16384 kB
Swap: total: 0 kB, used: 0 kB, free: 0 kB, shared: 0 kB, buff/cache: 0 kB, available: 0 kB

divo@MSI:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        4.0G  1.0G  3.0G  25% /
/dev/sda2        4.0G  1.0G  3.0G  25% /mnt/usb
/dev/sda3        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda4        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda5        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda6        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda7        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda8        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda9        4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda10       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda11       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda12       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda13       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda14       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda15       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda16       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda17       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda18       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda19       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda20       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda21       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda22       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda23       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda24       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda25       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda26       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda27       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda28       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda29       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda30       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda31       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda32       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda33       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda34       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda35       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda36       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda37       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda38       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda39       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda40       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda41       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda42       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda43       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda44       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda45       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda46       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda47       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda48       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda49       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda50       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda51       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda52       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda53       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda54       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda55       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda56       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda57       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda58       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda59       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda60       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda61       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda62       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda63       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda64       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda65       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda66       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda67       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda68       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda69       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda70       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda71       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda72       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda73       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda74       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda75       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda76       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda77       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda78       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda79       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda80       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda81       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda82       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda83       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda84       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda85       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda86       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda87       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda88       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda89       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda90       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda91       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda92       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda93       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda94       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda95       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda96       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda97       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda98       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda99       4.0G  1.0G  3.0G  25% /mnt/usb/lib
/dev/sda100      4.0G  1.0G  3.0G  25% /mnt/usb/lib
```